Exercise 13

!! Consider the calculations not relevant for the exam !

Problem 22

a) The fundamental soliton can propagate in a fiber without changing its shape. This is possible due to the fact that the dispersion (CD) and Kerr-nonlinearity (self-phase modulation) set up a robust balance.

b) When losing power during propagation the balance between CD and Kerr-nonlinearity is disturbed, the pulse starts to broaden, the Kerr-Effect can't compensate anymore for the CD. -> Problems in signal detection.

c) Two main Problems:
I) an increased pulse width
II) decreased power level that leads to low SNR. These problems were typically compensated by dispersion compensated fibers (DCF) and EDFAs.

Problem 23

a) FWM Phase Missmatch:

$$\Delta\beta = \beta(f_4) + \beta(f_3) - \beta(f_1) - \beta(f_2)$$

for FWM process: $f_4 = f_1 + f_2 - f_3$

Taylor expansion around center frequency f_0

$$\beta(f) = \beta_0 + \beta_1 (f - f_0) + \frac{1}{2}\beta_2 (f - f_0)^2 + \frac{1}{6}\beta_3 (f - f_0)^3$$

for small deviation, second term stays dominant:

$$\Delta\beta \approx \frac{1}{2}\beta_2 \left((f_4 - f_0)^2 + (f_3 - f_0)^2 - (f_1 - f_0)^2 - (f_2 - f_0)^2 \right)$$

-> convert λ to frequency

$f_1 = c/\lambda_1 = 193,548$ THz

$f_2 = 193,448$ THz

$f_3 = 193,348$ THz

$f_4 = f_1 + f_2 - f_3 = 193,648$ THz $\rightarrow \lambda_4 = 1549,2$ nm

Channel spacing: $\Delta f = 100\,GHz$

relate β_2 to dispersion D

$$\beta_2 = -\frac{D\lambda^2}{2\pi c} = \frac{-0.5\,\frac{ps}{nm\,km}\cdot(1.55\cdot10^{-6}\,m)^2}{2\pi\cdot3\cdot10^8\,\frac{m}{s}} = -0.635\,\frac{ps^2}{km}$$

$$\Delta\beta = \beta_2(\Delta\omega)^2 = \beta_2(2\pi\Delta f)^2$$

$\Downarrow$

Phase mismatch

$$\Delta\beta = -0.635\,\frac{ps^2}{km}\cdot(2\pi\cdot100\,GHz)^2$$

$$= -2.515\cdot10^{-10}\,rad/m$$

$$\Delta\beta\cdot L = -2.515\cdot10^{-10}\,rad/m\cdot20\cdot10^3\,m = -5.03\cdot10^{-6}\,rad \rightarrow \text{very good phase matching!}$$

b) $$\eta_{FWM} \approx (D_d\gamma\cdot L_{eff})^2\cdot\frac{\alpha^2}{\alpha^2+(\Delta\beta)^2}$$

with $D_d=6$ as the degeneracy factor: Degenerate: $2\omega_1-\omega_2\rightarrow\omega_3$ $D_d=2$

$\omega_1+\omega_1-\omega_2\rightarrow\omega_3$ $D_d=2$

$\omega_1+\omega_2-\omega_3\rightarrow\omega_4$ $D_d=6$

$\omega_1+\omega_1-\omega_1\rightarrow\omega_1$ $D_d=1$

$$\alpha = \frac{\alpha_{dB}\cdot\ln(10)}{10} = \frac{0.2\cdot\ln(10)}{10} = 0.046\,\frac{1}{km}$$

$$\eta_{FWM} = \left(6\cdot2\,\frac{1}{km}\cdot20\,km\right)^2\cdot\frac{\left(0.046\,\frac{1}{km}\right)^2}{\left(0.046\,\frac{1}{km}\right)^2+\left(-2.515\cdot10^{-7}\,\frac{rad}{km}\right)}$$

$$\eta_{FWM} = 57.600$$

$$P_{FWM} = \eta_{FWM}\cdot P_1\cdot P_2\cdot P_3 = 57600\cdot0.01\,mW\cdot0.01\,mW\cdot0.01\,mW = 57.6\,mW$$

$\hookrightarrow 17.6\,dBm$

c) Maximum FWM Efficiency wavelength

Max FWM at zero Dispersion wavelength! where $\Delta\beta=0$

$$\lambda_{z0} = \lambda_0 - \frac{D}{S} = 1550 - \frac{0.5}{0.05} = 1540\,nm$$

Problem 24

a) Number of FWM products:

degenerate FWM: $(f_i + f_i - f_1)$: $N(N-1)$ products

for $N = 16$ $16 \times 15 = 240$ products

Non-degenerate FWM: $(f_i + f_j - f_k,\ i \neq j \neq k) = N(N-1)(N-2)/2$ products

$N = 16$ $16 \cdot 15 \cdot 14 / 2 = 1680$ products

↳ total FWM products: 1920

Approximate formula for in band products: $N_{in\text{-}band} = \frac{N^3}{4} = 1024$

b) worst case crosstalk: -> when multiple FWM products overlap!

⟶ calculation here is quite complex, i decided to leave it out from the solution.

c) -> reduce channel power

-> increase channel spacing

-> use different polarisations.